

# SAMAY THAKUR

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## EDUCATION

### M.S. Ramaiah Institute of Technology

Bachelor of Engineering (B.E.) | Electronics and Instrumentation Engineering | CGPA: 7.95

Oct 2023 – Present

Bengaluru, India

### Sunbeam School Varuna

Grade X (CBSE): 89.6% | Grade XII (CBSE): 74.2%

Mar 2019 – Jun 2022

Varanasi, India

## EXPERIENCE

### CPS & IoT Intern

Aug 2024 – Aug 2024

Intra Institutional Internship, MSRIT

Bengaluru, India

- Architected a real-time water-level monitoring and automated pump-control system on Arduino Uno using ultrasonic sensors; implemented threshold-based control logic in C++ with sensor data acquisition pipeline.
- Validated system reliability across edge-case water-level conditions, eliminating tank overflow and dry-run failure modes through automated ON/OFF pump control.

## PROJECTS

### CynLr EPM PoC | GitHub

- Owned end-to-end program scoping for a robotics PoC spanning moving-object tracking, track-and-grasp, hardware integration, and delivery - structured 41 top-level workstreams and 97 subtasks across robotics, CV/ML, software, mechanical, electronics, procurement, and external vendors.
- Built a 604-person-hour effort model in Excel for resource planning across all workstreams; mapped dependencies, milestones, risks, and external constraints into an execution framework in Asana.
- Defined a KPI and progress-monitoring framework across 5 dimensions - dependency resolution, development progress, infrastructure readiness, testing outcomes, and delivery readiness - enabling objective milestone tracking.
- Tools/Technologies: Asana, Microsoft Excel

### SmartQC: On-Device Visual Quality Inspection | GitHub

Qualcomm | Snapdragon® AI Lab Build & Present Challenge

- Authored a PRD defining user problem, target segment, success metrics (field accuracy  $\geq 97\%$ ), and feature scope - scoped and shipped a 0→1 on-device visual QC product targeting small food producers priced out of industrial inspection hardware, with all cut decisions documented before writing a single line of code.
- Drove model performance to 99.2% validation accuracy (0.9997 ROC-AUC) on Qualcomm AI Hub for Snapdragon; stress-tested against 352 uncured real-world images, achieving 97.4% field accuracy — a gap that directly informed a confidence-threshold feature for low-confidence prediction flagging.
- Prioritized three shipped features — explainability, batch processing, confidence flagging — against user workflow needs, explicitly cutting scope and documenting trade-offs in a technical report for reproducibility.
- Tools/Technologies: Python, PyTorch, ONNX, Qualcomm AI Hub, PyQt5, OpenCV, Scikit-learn

### AI-Based Bread Quality Indicator System | GitHub

- Defined product scope for an end-to-end bread-quality assessment system combining gas-sensing (MQ-135) and visual inspection (ESP32-CAM) - eliminating dependency on single-modality detection.
- Shipped a MobileNetV2 image classification pipeline with 128×128 preprocessing to 100% validation accuracy across 15 training epochs; deployed model inference on-device via TensorFlow Lite.
- Integrated real-time Wi-Fi telemetry, automated LED/buzzer alerts, and model predictions into a Flask monitoring dashboard — reducing time-to-alert for quality failures to sub-second.
- Tools/Technologies: Python, TensorFlow, MobileNetV2, TensorFlow Lite, Flask, ESP32-CAM

### Stock Market Price Prediction & Analysis | GitHub

- Framed next-day NIFTY 50 price forecasting as a regression problem; engineered 4 predictive feature families - RSI, MACD, Moving Averages, Bollinger Bands - from raw historical market data.
- Achieved 1.39% MAPE and 93%+  $R^2$  against baseline forecasting models, validating feature engineering approach.
- Tools/Technologies: Python, TensorFlow, LSTM, Pandas, NumPy, Scikit-learn

## TECHNICAL SKILLS

**PM & Program Tools:** Asana, Microsoft Excel, Program Planning, Resource Planning, Dependency Mapping, Milestone Tracking, KPI Tracking, Risk Identification, Feature Prioritization, PRD Authoring, Cross Functional Coordination

**AI/ML:** PyTorch, TensorFlow, TensorFlow Lite, Scikit-learn, Computer Vision, Deep Learning, ONNX, Qualcomm AI Hub

**Data & Analytics:** SQL, Excel, Tableau, Power BI, A/B Testing Fundamentals

**Engineering:** Python, C++, OpenCV, Flask, PyQt5, SQLite, Git, GitHub, VS Code, Jupyter

**Concepts:** Algorithms, Data Structures, DBMS, Operating System, Computer Networks, Digital Logic Design

## ACHIEVEMENTS & LEADERSHIP

- Student Member:** Instrument Society of India (ISOI), IISc Bangalore (Nov 2023 – Oct 2027)
- Community Volunteer:** Organized and supported blood-donation drives through school and community initiatives, facilitating donor engagement and community welfare.